

# An Empirical Study of Representation Learning for Reinforcement Learning in Healthcare

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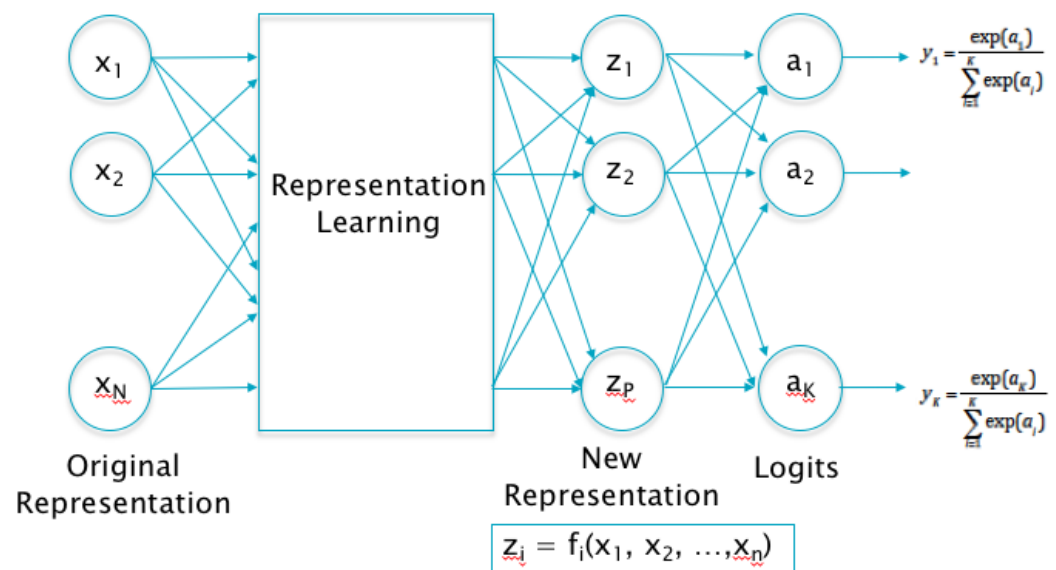
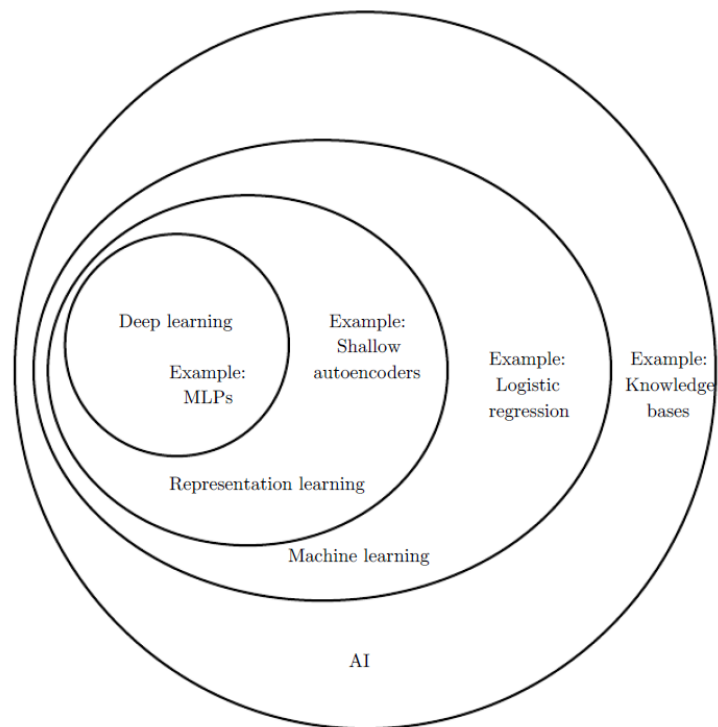
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석사과정

김홍범

# 1. Introduction

## 1. What is Representation Learning?



### Representation Learning

- 최종 task의 유형에 따라 '**New Representation**'에 해당하는 **new feature**(<--->input feature)을 뽑게 학습하는 방법
- supervised든 unsupervised든 모두 objective(loss) function이 존재하므로 new representation에 해당하는 hidden feature vector를 구할 수 있게 됨

# 2. Paper Review

## Introduction

### An Empirical Study of Representation Learning for Reinforcement Learning in Healthcare

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#### Abstract

Reinforcement Learning (RL) has recently been applied to sequential estimation and prediction problems identifying and developing hypothetical treatment strategies for septic patients, with a particular focus on *offline* learning with observational data. In practice, successful RL relies on informative latent states derived from sequential observations to develop optimal treatment strategies. To date, how best to construct such states in a healthcare setting is an open question. In this paper, we perform an empirical study of several information encoding architectures using data from septic patients in the MIMIC-III dataset to form representations of a patient state. We evaluate the impact of representation dimension, correlations with established acuity scores, and the treatment policies derived from them. We find that sequentially formed state representations facilitate effective policy learning in batch settings, validating a more thoughtful approach to representation learning that remains faithful to the sequential and partial nature of healthcare data.

**Keywords:** representation learning, reinforcement learning, partial observability, sequential autoencoding

#### 1. Introduction

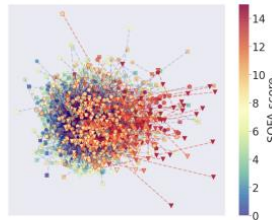


Figure 1: The first and final observations of septic patients in MIMIC-III, colored by SOFA score, visualized via Principal Component Analysis. Blue lines connect observations of patients who recovered, while red lines signify those that did not. Notably, these raw observations of severe health are not directly separable.

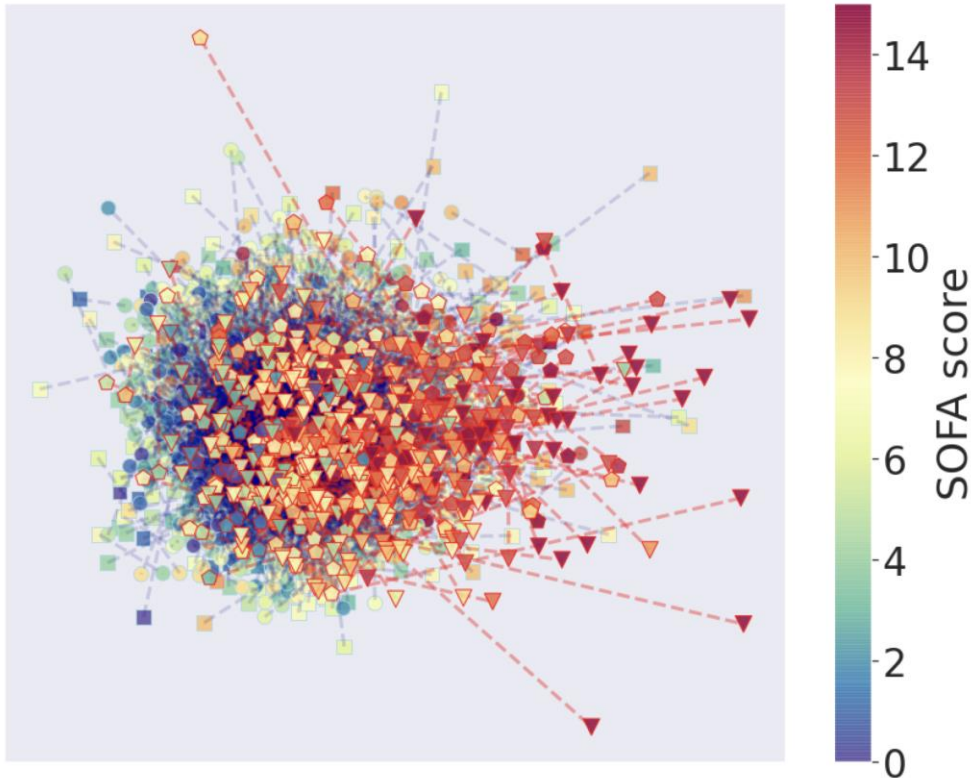
Many problems in healthcare are a form of sequential decision making, e.g., clinical staff making decisions about the best “next step” in care (Ghassemi et al., 2019). Solving these problems is similar to finding an optimal decision making policy, requiring estimation and optimization of the cumulative effects of decisions over time (Sox et al., 2007). Recently, reinforcement learning (RL) has been proposed as a promising approach for finding an optimal policy for such processes from data (Gottes-

- 정식 ACCEPT되진 않은 PAPER
- But 다양한 방식의 representation methods를 health care데이터에 사용했다는 점에서 의의가 있음

## 2. Paper Review

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### Why representation learning?



Colored by SOFA score, visualized via PCA

Datasets : MIMMIC- III

- RL relies on informative latent states
- These raw observation of severe health are not directly separable

Objective

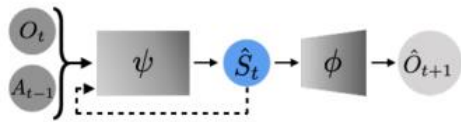
1. Effect of representation dimension
2. Impact of including Contextual information(5 demographic features)
3. Find treatment policies from off-line RL algorithm

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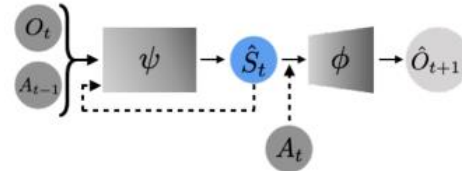
### Methods

| Approach | Recurrent | Sequence as input | Num. Parameters |
|----------|-----------|-------------------|-----------------|
| AE       |           |                   | 27k – 76k       |
| AIS      | ✗         |                   | 28k – 339k      |
| CDE      | ✗         | ✗                 | 78.9k – 1.78m   |
| DDM      | ✗         |                   | 6k – 1.25m      |
| DST      | ✗         | ✗                 | 47k – 256k      |
| ODE      | ✗         |                   | 48.3k – 329k    |
| RNN      | ✗         |                   | 26k – 337k      |

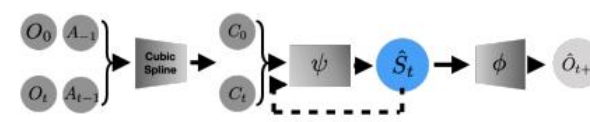
a) RNN



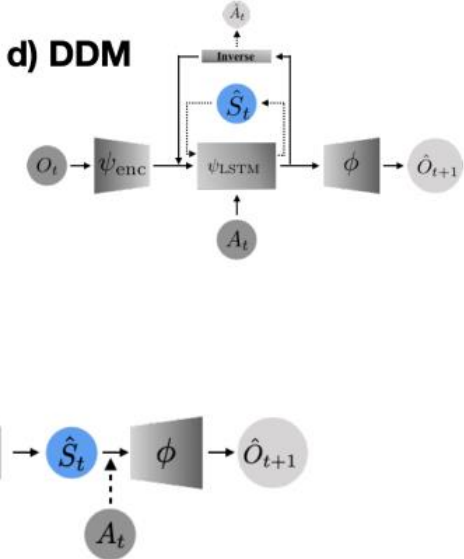
b) AIS



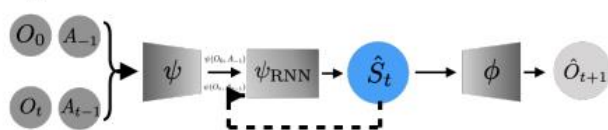
c) CDE



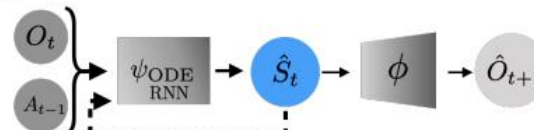
d) DDM



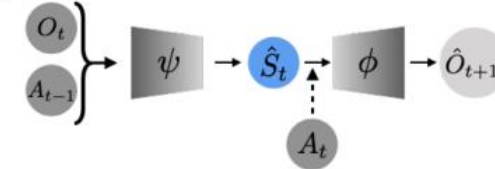
e) DST



f) ODE



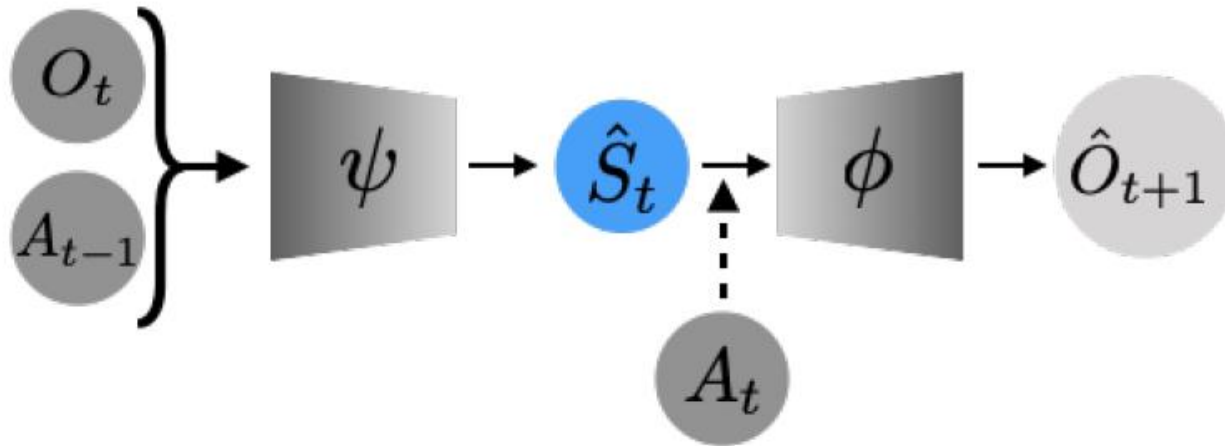
g) AE



Six recurrent model + simple non-recurrent Autoencoder

## 2. Paper Review

### Methods



$O_t$  : Observation

$A_{t-1}$  : Treatment Action

$\hat{S}_t$  : State Representation

$\psi$  : Encoding function(model)

$\phi$  : Decoding function(model)

### Autoencoder architecture

$$LOSS = L(O_{t+1}, \hat{O}_{t+1}) - \lambda \rho(\hat{S}_t)$$

$$\lambda \rho(\hat{S}_t) = \lambda_1 \rho^{SOFA}(\hat{S}_t) + \lambda_2 \rho^{SAPS II}(\hat{S}_t) + \lambda_3 \rho^{OASIS}(\hat{S}_t) \quad \rightarrow$$

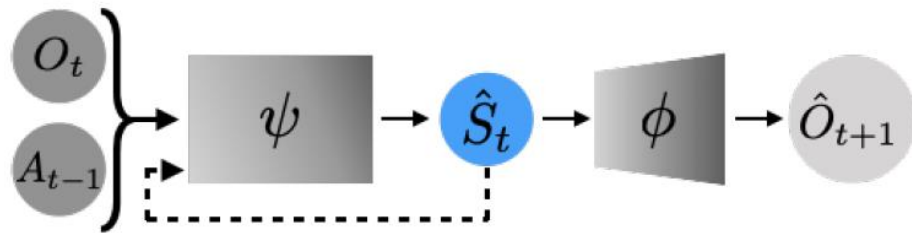
Regularization term

SOFA, SAPS II, OASIS : Independent acuity score

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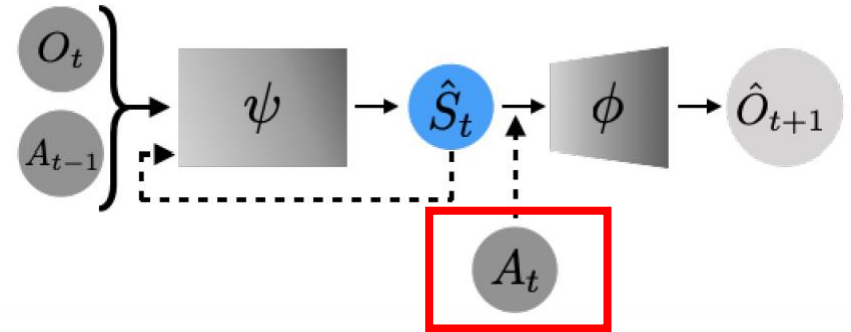
### Methods

Basic RNN architecture



$\psi$  : 3-layer Recurrent Neural Network  
 $\phi$  : 3-layer feed-forward neural network

Approximate Information State(AIS)

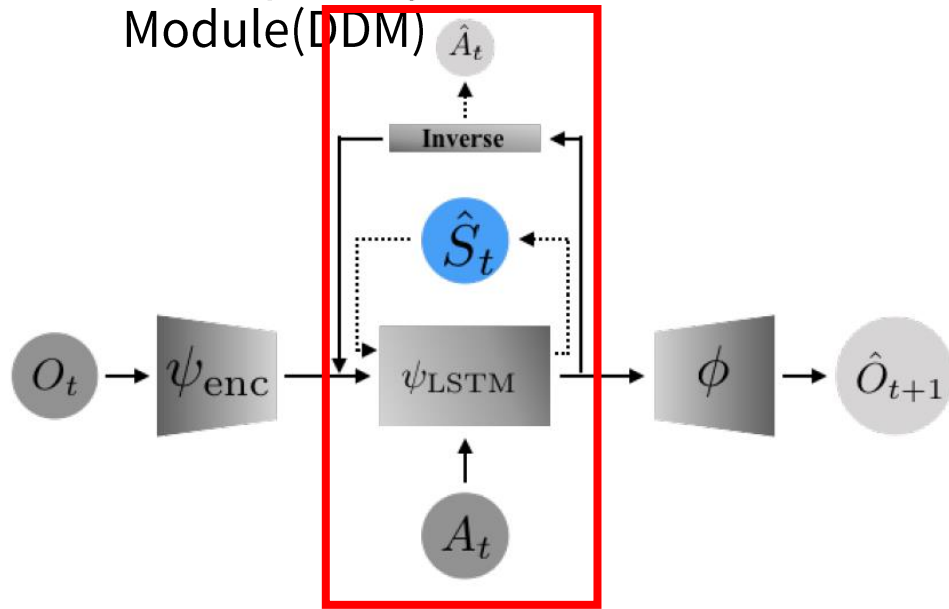


$\psi$  : gated recurrent unit(GRU)  
 $\phi$  : 3-layer feed-forward neural network  
 $A_t$  : Current action

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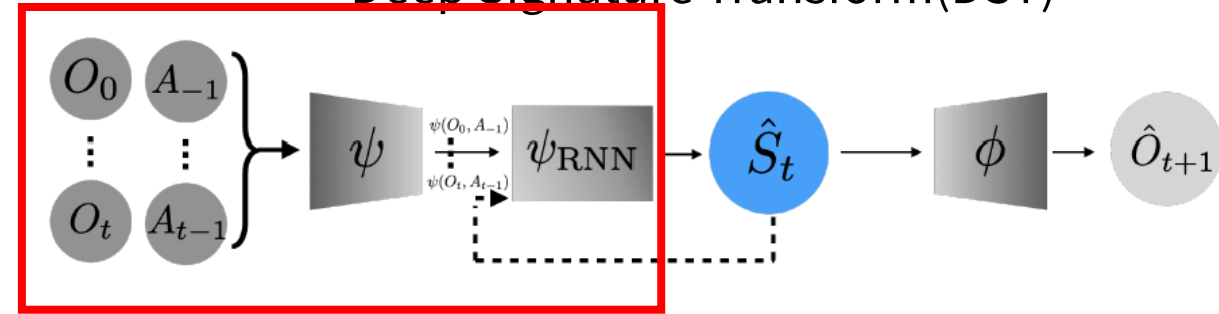
### Methods

Decoupled Dynamics  
Module(DDM)



$\psi_{\text{enc}}$  : 3-layer feed-forward neural network(ELU)  
 $\psi_{LSTM}$  : two linear layers, LSTM Cell  
 $\phi$  : 3-layer feed-forward neural network

Deep Signature Transform(DST)



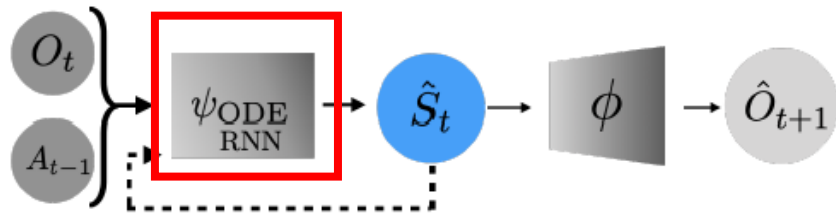
$\psi$  : pointwise encoder  
 $\psi_{\text{enc}}$  : 3-layer feed-forward neural network(ELU)  
 $\psi_{LSTM}$  : two linear layers, LSTM Cell  
 $\phi$  : 3-layer feed-forward neural network



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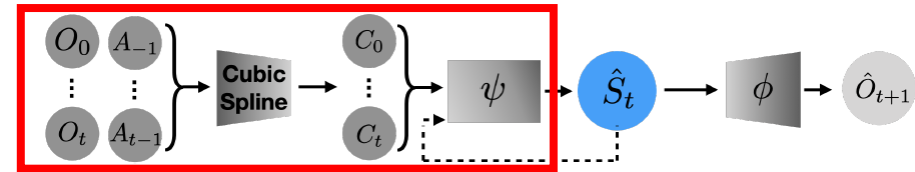
### Methods

ODE RNN



$\psi_{ODE\_RNN}$  : ODE\_RNN  
 $\phi$  : 3-layer feed-forward neural network

Neural Control Differential Equations(CDEs)

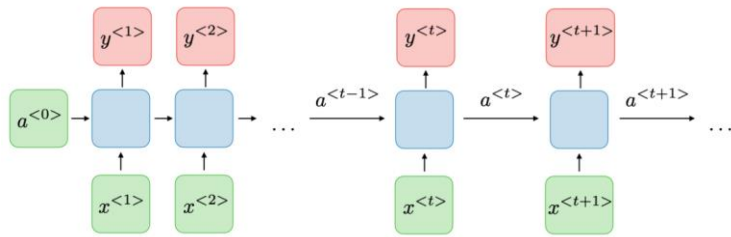


*Cubic Spine* : observation, action  $\Delta t$ 의 coefficient  
 $\psi$  : MLP(100 layer)  
 $\phi$  : 3-layer feed-forward neural network

## 2. Paper Review

### Methods – ODE Network

RNN 구조



Ordinary Differential Equations  
(미분방정식)

$$\vec{y}_{i+1} = \begin{pmatrix} y_{i+1} \\ \vdots \\ y_{i+1}^{(N-1)} \\ y_{i+1}^{(N)} \end{pmatrix} = \begin{pmatrix} y_i + h \cdot y'_i \\ \vdots \\ y_i^{(N-1)} + h \cdot y_i^{(N)} \\ y_i^{(N)} + h \cdot f(t_i, y_i, y'_i, \dots, y_i^{(N)}) \end{pmatrix}$$

$$\mathbf{a}_t = g_1(W_{aa}\mathbf{a}_{t-1} + W_{ax}\mathbf{x}_t + \mathbf{b}_a)$$

$$\mathbf{y}_t = g_2(W_{ya}\mathbf{a}_t + \mathbf{b}_y)$$

동일 가정(Autoregressive)

$$\begin{bmatrix} \mathbf{a}_t \\ \mathbf{x}_{t+1} \end{bmatrix} = \begin{bmatrix} \mathbf{a}_{t-1} \\ \mathbf{x}_t \end{bmatrix} + F \left( \begin{bmatrix} \mathbf{a}_{t-1} \\ \mathbf{x}_t \end{bmatrix} \right)$$

$g_1, g_2$  : activation function  
 $\mathbf{x}_t$  : input  
 $\mathbf{a}_t$  : hidden state  
 $\mathbf{y}_t$  : output

Euler  
methods

$$F \left( \begin{bmatrix} \mathbf{a} \\ \mathbf{x} \end{bmatrix} \right) = \begin{bmatrix} g_1(W_{aa}\mathbf{a} + W_{ax}\mathbf{x} + \mathbf{b}_a) - \mathbf{a} \\ g_2(W_{ya}g_1(W_{aa}\mathbf{a} + W_{ax}\mathbf{x} + \mathbf{b}_a) + \mathbf{b}_y) - \mathbf{x} \end{bmatrix}$$

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### Methods – ODE Network

Q :위와 같이 ODE Network를 만들면, inference는 미분방정식을 풀 수 있다. 그러면 Backprop, 즉 모델의 parameter update는 어떻게 해야 하는 걸까요?

A : Loss 를 계산하고 backprop하기 위해서 역순으로 새로운 ODE를 계산하고 이를 기존의 ODE값과 비교하고 업데이트를 진행해 나감

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#### Algorithm 1 Reverse-mode derivative of an ODE initial value problem

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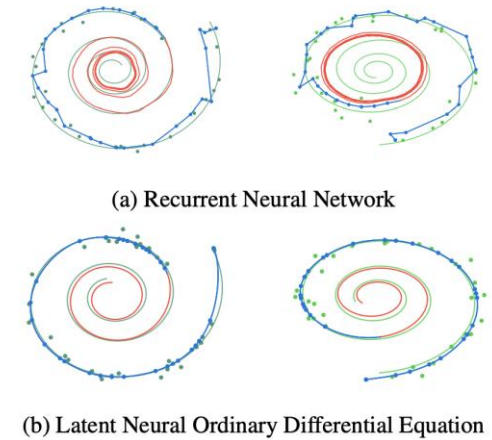
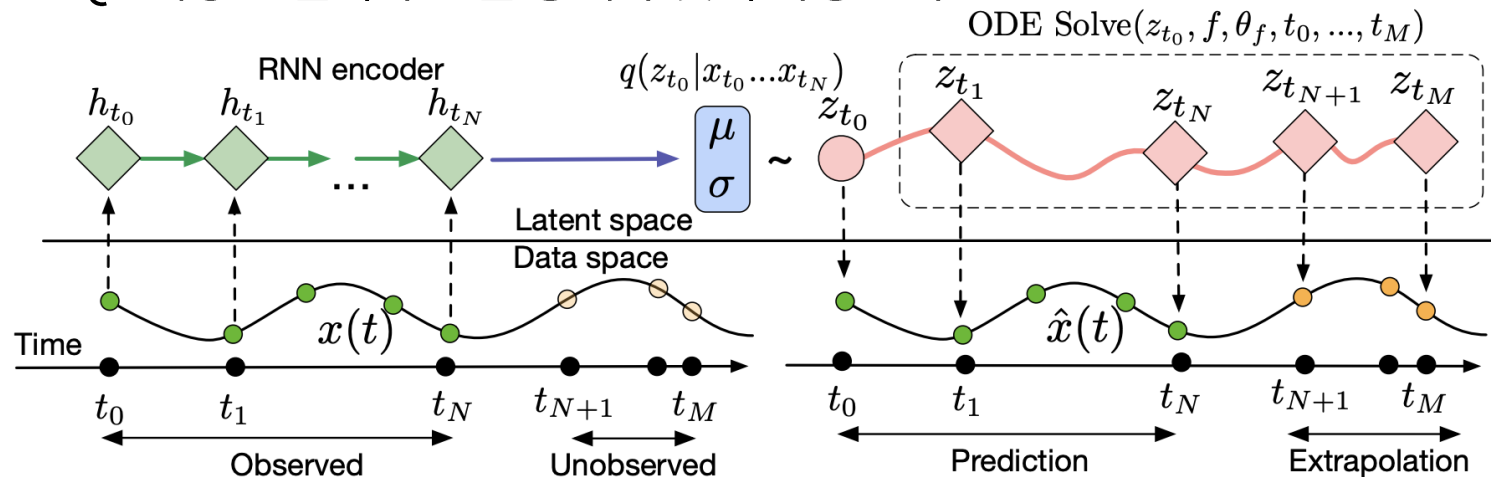
**Input:** dynamics parameters  $\theta$ , start time  $t_0$ , stop time  $t_1$ , final state  $\mathbf{z}(t_1)$ , loss gradient  $\partial L / \partial \mathbf{z}(t_1)$   
 $s_0 = [\mathbf{z}(t_1), \frac{\partial L}{\partial \mathbf{z}(t_1)}, \mathbf{0}_{|\theta|}]$  ▷ Define initial augmented state  
**def** aug\_dynamics( $[\mathbf{z}(t), \mathbf{a}(t), \cdot], t, \theta$ ): ▷ Define dynamics on augmented state  
    **return**  $[f(\mathbf{z}(t), t, \theta), -\mathbf{a}(t)^\top \frac{\partial f}{\partial \mathbf{z}}, -\mathbf{a}(t)^\top \frac{\partial f}{\partial \theta}]$  ▷ Compute vector-Jacobian products  
 $[\mathbf{z}(t_0), \frac{\partial L}{\partial \mathbf{z}(t_0)}, \frac{\partial L}{\partial \theta}] = \text{ODESolve}(s_0, \text{aug\_dynamics}, t_1, t_0, \theta)$  ▷ Solve reverse-time ODE  
**return**  $\frac{\partial L}{\partial \mathbf{z}(t_0)}, \frac{\partial L}{\partial \theta}$  ▷ Return gradients

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## 2. Paper Review

### Methods – ODE Network

Q : 해당 모델의 구조를 통해 무엇이 가능한지?



ODE의 초기 Timestamp에서의 몇 개의 값을 바탕으로 이후의 값을 예측하는 task가 가능함  
기존의 RNN에 비해 RMSE 값이 더 작음

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### Reinforcement Learning

BCQ(Batch Constrained Q-Learning)

-RL Algorithm

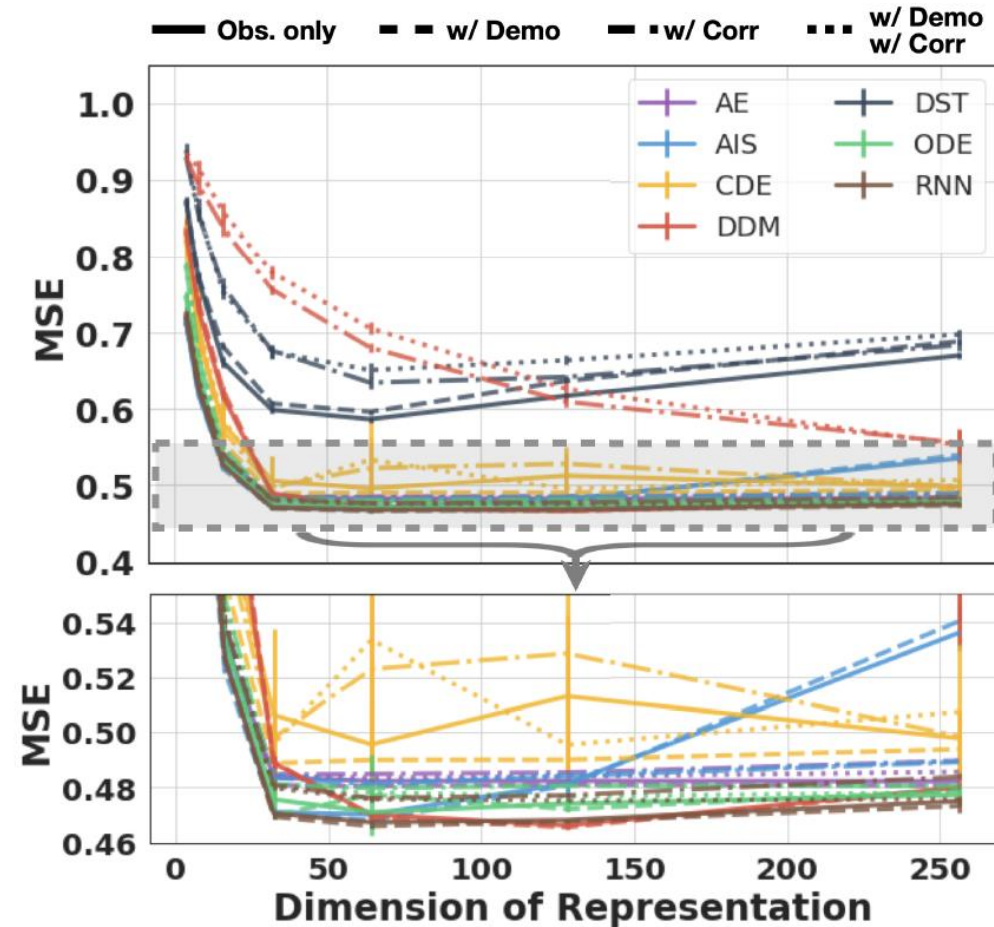
WIS(weighted importance sampling)

- Testing subset

| Approach | Best MSE           | $\hat{d}_s$ | Training Setting               |
|----------|--------------------|-------------|--------------------------------|
| AE       | $0.4804 \pm 0.001$ | 64          | w/ $\mathcal{O} + \mathcal{D}$ |
| AIS      | $0.4679 \pm 0.004$ | 64          | $\mathcal{O} + \mathcal{D}$    |
| CDE      | $0.4887 \pm 0.019$ | 32          | $\mathcal{O} + \mathcal{D}$    |
| DDM      | $0.4654 \pm 0.002$ | 128         | $\mathcal{O} + \mathcal{D}$    |
| DST      | $0.5863 \pm 0.013$ | 64          | $\mathcal{O}$                  |
| ODE      | $0.4698 \pm 0.003$ | 32          | $\mathcal{O} + \mathcal{D}$    |
| RNN      | $0.4658 \pm 0.002$ | 64          | $\mathcal{O} + \mathcal{D}$    |

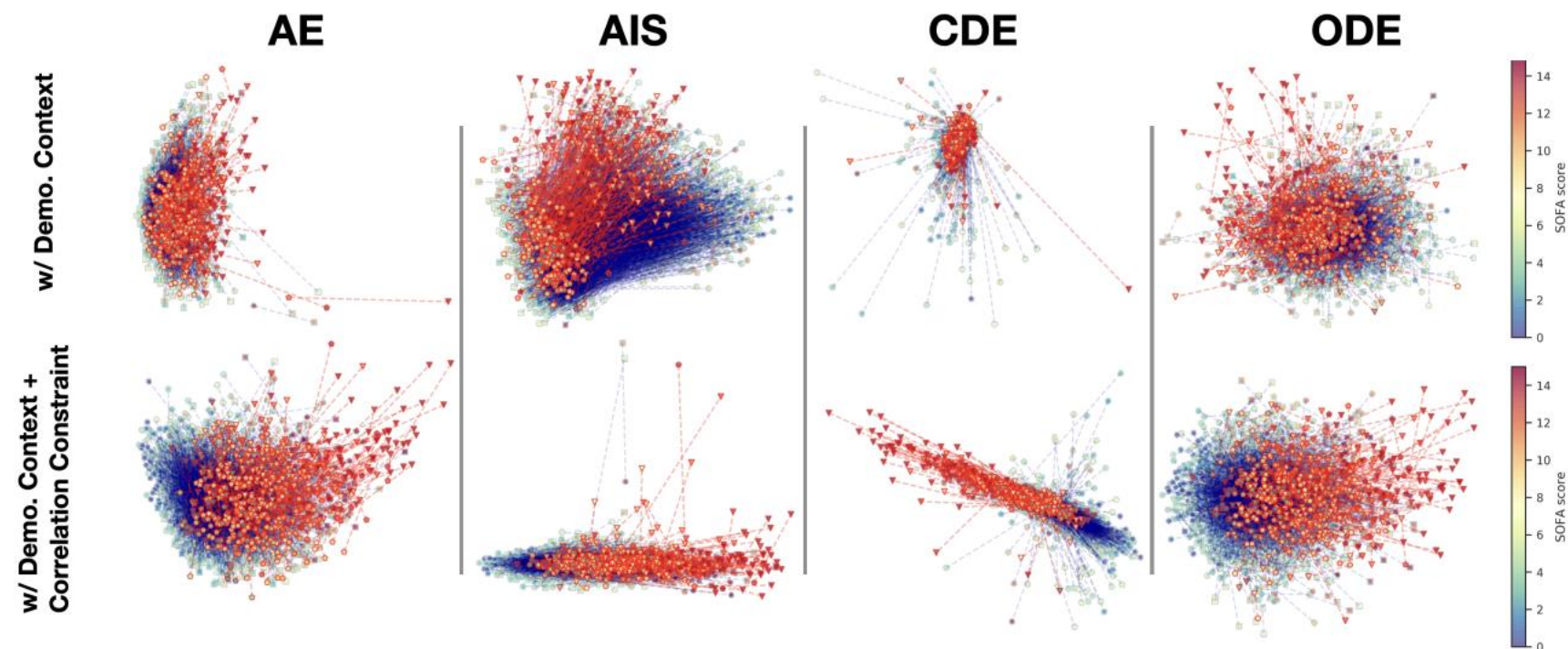
O : Observation  $\mathcal{D}$  : Demographics

Highest capacity representation may not be the most informative!



## 2. Paper Review

### Results – latent PCA





## 2. Paper Review

### Results – latent PCA

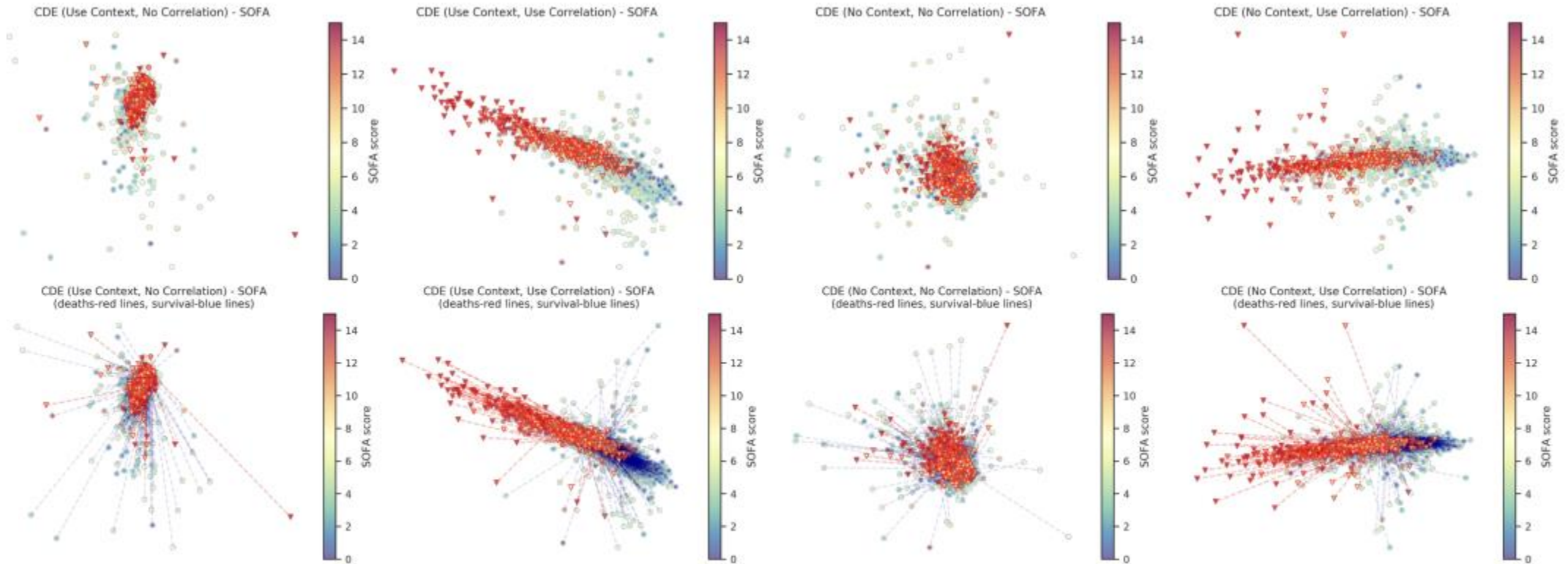


Figure 18: Representations of patient health, learned through the Neural CDE (CDE)

# 3. Conclusion

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## Results

- 다양한 representation 기법을 통하여 health care 데이터에 적용하고 이를 바탕으로 RL 모델을 도출함

## Limitation

구체적인 RL에 따른 Policy의 해석이 부족함

Action 및 reward의 정의가 부족함

최종적으로는 state representation의 다양한 기법의 소개라는 한계점을 지님

## 연구 활용방안

- 현재 연구중인 PPMI(파킨슨병) 데이터셋은 Multimodal dataset이고 특유의 라벨 적용이 어려움
- 해당 논문에서 제안하는 다양한 state representation 기법을 적용하는 중
- 이를 이용하여 향후 RL(강화학습)에서 사용할 STATE를 Define할 예정
- 더 나아가 이를 이용한 구체적인 투약 policy 수립 및 평가를 진행하고 임상적 의미 파악 예정



# THANK YOU

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